COMP 441 — Software Analysis and Testing

Week 8 — TDD Cycles and BDD Acceptance Scenarios

Group A · Track D · TOML Kit

Period	Week 8
Evidence folder	week8
Repository	github.com/the-dezeray/tomlkit
Commit(s)	a4f54db / a38a3d8 / c829a38
Generated	2026-10-27
Contents	13 sections

This pack collects the week's human-readable report and key evidence. Raw AI-only logs are listed in the inventory and intentionally not printed.

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Source: week8\TDD_LOG.md

TDD log (Week 8, branch `week8-tdd`)

Each cycle has three commits: **red** (failing test only), **green** (minimal code change), **refactor** (no behaviour change, full suite green). Raw pytest output for each step is in `tdd_runs/`. Times are commit times (SAST, UTC+2).

Cycle	Defect	Step	Commit	Time	Evidence
C1	D1 `date()` accepts datetime strings	Red	`9472f8a`	2026-10-18 23:00:55	`tdd_runs/c1_red.txt`: 4 failed, 1 passed
		Green	`a4f54db`	23:01:24	one-line change in `api.date`; D1 xfail markers removed; 1,475 passed
		Refactor	`fad6c12`	23:01:45	`_parse_rfc3339_as()` shared by date/time/datetime; 1,475 passed
C2	D7 multiline strings emit a bare CR	Red	`51c6863`	23:01:59	`tdd_runs/c2_red.txt`: 6 failed, 2 passed
		Green	`a38a3d8`	23:03:22	`String.from_raw` escapes (MLB) or rejects (MLL) a lone CR; 1,485 passed
		Refactor	`26421d6`	23:03:40	regex named once as `_BARE_CR`; 1,485 passed
C3	D6 `comment()` with a bare CR	Red	`3c33afc`	23:03:51	`tdd_runs/c3_red.txt`: 2 failed, 2 passed
		Green	`c829a38`	23:04:11	`api.comment` treats a lone CR as a line break; 1,489 passed
		Refactor	`b60559f`	23:04:37	`BARE_CR` moved to `_utils` and shared by C2 and C3; 1,489 passed

Tests: `tests_tdd/test_tdd_cycles.py`. Full-suite count = tests/ + tests_ai/ + tests_manual/ + tests_property/ + tests_tdd/.

C1 red — date() accepts datetime strings

Source: week8\tdd_runs\c1_red.txt

```
=========================== short test summary info ============================
FAILED tests_tdd/test_tdd_cycles.py::test_c1_date_rejects_datetime_strings[1979-05-27T07:32:00]
FAILED tests_tdd/test_tdd_cycles.py::test_c1_date_rejects_datetime_strings[1979-05-27 07:32:00Z]
FAILED tests_tdd/test_tdd_cycles.py::test_c1_date_rejects_datetime_strings[0001-10-30 00:00]
FAILED tests_tdd/test_tdd_cycles.py::test_c1_date_rejects_datetime_strings[1979-05-27T07:32:00-08:00]
4 failed, 1 passed in 0.20s
```

C1 green

Source: week8\tdd_runs\c1_green.txt

```
5 passed in 0.23s
1475 passed, 5 xfailed in 7.62s
```

C1 refactor

Source: week8\tdd_runs\c1_refactor.txt

```
1475 passed, 5 xfailed in 7.57s
```

C2 red — bare CR in multiline strings

Source: week8\tdd_runs\c2_red.txt

```
=========================== short test summary info ============================
FAILED tests_tdd/test_tdd_cycles.py::test_c2_multiline_basic_escapes_bare_cr[a\rb]
FAILED tests_tdd/test_tdd_cycles.py::test_c2_multiline_basic_escapes_bare_cr[\r]
FAILED tests_tdd/test_tdd_cycles.py::test_c2_multiline_basic_escapes_bare_cr[a\r\rb]
FAILED tests_tdd/test_tdd_cycles.py::test_c2_multiline_basic_escapes_bare_cr[end\r]
FAILED tests_tdd/test_tdd_cycles.py::test_c2_multiline_literal_rejects_bare_cr[a\rb]
FAILED tests_tdd/test_tdd_cycles.py::test_c2_multiline_literal_rejects_bare_cr[\r]
6 failed, 2 passed, 5 deselected in 0.19s
```

C2 green

Source: week8\tdd_runs\c2_green.txt

```
13 passed in 0.29s
1485 passed, 3 xfailed in 6.79s
```

C2 refactor

Source: week8\tdd_runs\c2_refactor.txt

```
1485 passed, 3 xfailed in 6.79s
```

C3 red — comment() with a bare CR

Source: week8\tdd_runs\c3_red.txt

```
========================== short test summary info ===========================
FAILED tests_tdd/test_tdd_cycles.py::test_c3_comment_splits_every_line_break[a\rb-# a\n# b]
FAILED tests_tdd/test_tdd_cycles.py::test_c3_comment_splits_every_line_break[a\r\rb-# a\n#\n# b]
2 failed, 2 passed, 13 deselected in 0.17s
```

C3 green

Source: week8\tdd_runs\c3_green.txt

```
17 passed in 0.17s
1489 passed, 3 xfailed in 6.63s
```

C3 refactor

Source: week8\tdd_runs\c3_refactor.txt

```
1489 passed, 3 xfailed in 6.90s
```

BDD feature — edit config and preserve layout

Source: tests_bdd\features\edit_config.feature

```
Feature: Edit a project config file without losing its formatting
  As a Python developer who keeps tool settings in pyproject.toml
  I want to change one value through tomlkit
  So that my comments, spacing and key order survive the edit

  # Level: acceptance (end to end: real file on disk -> TOMLFile -> edit -> file on disk)
  # RTM: R03 (edits preserve untouched layout), R07 (file round-trip)
  Scenario: Bump the version and keep every comment and blank line
    Given a config file containing:
      """
      # Project settings
      [project]
      name = "demo"  # do not rename
      version = "1.4.2"

      # Tools
      [tool.pytest]
      addopts = "-q"
      """
    When I open the file with tomlkit and set "project.version" to "1.5.0"
    And I save the file
    Then the file on disk is exactly:
      """
      # Project settings
      [project]
      name = "demo"  # do not rename
      version = "1.5.0"

      # Tools
      [tool.pytest]
      addopts = "-q"
      """
    And reading "project.version" from the saved file gives "1.5.0"
```

BDD feature — backlog scenarios

Source: tests_bdd\features\backlog.feature

```
# Drafted in Week 8, to be automated in Week 9 (not yet bound to step definitions).
Feature: Work with real-world TOML files safely

  # Level: acceptance
  # RTM: R04 (malformed input reports line and column)
  Scenario: A typo in a config file is reported at the right place
    Given a config file whose third line is "version = 1.4.2"
    When I try to open it with tomlkit
    Then I get a parse error that names line 3
    And the file on disk is unchanged

  # Level: acceptance
  # RTM: R07 (TOMLFile preserves Windows line endings)
  Scenario: Editing a Windows file keeps CRLF line endings
    Given a config file saved with CRLF line endings
    When I add the key "tool.ruff.line-length" with value 100 and save
    Then every line in the file ends with CRLF
    And the new key reads back as 100
```

The BDD step implementations are `tests_bdd/test_edit_config.py` and `tests_bdd/test_backlog.py`. The full red/green/refactor logs are in `week8/tdd_runs/`.

Raw evidence inventory

The pack prints the readable report and key summaries. Large raw artefacts are kept in the repository and listed here so a marker can verify them without paging through thousands of lines. Items tagged **[AI-only]** are deliberately not printed: they are the AI baseline's raw output, not the final hybrid result.

File	Size	Tag
week8\TDD_LOG.md	1 KB	
week8\tdd_runs\c1_green.txt	52 B	
week8\tdd_runs\c1_red.txt	503 B	
week8\tdd_runs\c1_refactor.txt	33 B	
week8\tdd_runs\c2_green.txt	53 B	
week8\tdd_runs\c2_red.txt	632 B	
week8\tdd_runs\c2_refactor.txt	33 B	
week8\tdd_runs\c3_green.txt	53 B	
week8\tdd_runs\c3_red.txt	317 B	
week8\tdd_runs\c3_refactor.txt	33 B	